

Advances in Algorithms (CS503)
Mid Semester Examination, September 2016
Department of CSE, IIT Patna

Total Marks: 60

Duration: 120 min

1. Answer the following questions: [5+2+8]

- I. Define Interval Scheduling problem. Present an optimal greedy algorithm for that.
- II. Define Independent set and Matroid.
- III. We have a collection $M = (S, I)$ made from an undirected graph $G = (V, E)$ where
 - a. $S = E$, the set of edges of G , and
 - b. If A is a subset of E , then $A \in I$ iff A is acyclic.Prove that M is a matroid.

2. Answer the following questions: [3 +4+8]

- I. When should we use Dynamic Programming and when should we go for Divide and Conquer?
- II. What are the differences between Top-down DP (Memoization) and Bottom-up DP (Tabulation)?
- III. Give a branching strategy and a bounding strategy for solving Travelling Salesman problem using Branch and Bound approach.

3. Answer the following questions: [2 +5+3+5]

- I. What is the main difference between Accounting method and Potential method in amortized analysis?
- II. Describe how to generate Residual graph. Why do we use augmenting path in solving Max-flow problem?
- III. How can we solve Bipartite matching problem from an algorithm for Max-flow problem?
- IV. What do we mean by intractable problem in practice and what are the strategies do we take to manage them?

4. Answer the following questions: [3+8+4]

- I. Define Approximation ratio, PTAS and FPTAS.
- II. Give a FPTAS for Subset-sum problem.
- III. What are the properties of Monte Carlo and Las Vegas algorithms? Give one example for each of them.